

PENGZHEN LI

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EDUCATION

University of Tennessee, Knoxville Ph.D. in Economics	expected 2027
Shandong University, Jinan M.A. in Finance — thesis published in <i>Technological Forecasting and Social Change</i> (2019); 1,000+ citations	2018
Tongji University, Shanghai B.A. in Economics	2015

RESEARCH INTERESTS

Environmental and Energy Economics, Agricultural and Resource Economics, Applied Econometrics, and International Trade.

PEER-REVIEWED PUBLICATIONS

Four SSCI/SCI-indexed publications; 1,132 total citations, h-index 4 (Google Scholar, August 2026).

“Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data.” K. Du, **P. Li**, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303. **1,000+ citations (Google Scholar)**.

“Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability.” **P. Li**, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

“Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA.” **P. Li**, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

“Beef price spread relationship with processing capacity utilization.” C. Martinez, **P. Li**, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

WORKING PAPERS

Measured Quality Falls Within the Firm: Market Access and Chinese Exports with James Lake [[Job Market Paper](#)]. [[paper](#)]

Abstract: Better access to a market is expected to make exporters upgrade. We show it does the opposite, and we locate where. Using the removal of tariff uncertainty for Chinese exporters to the United States in 2002, and comparing a firm’s shipments of a product to the United States against its shipments of the same product elsewhere, we find that measured export quality falls by about 14 percent for heavily exposed products against barely exposed ones, and that 78% of that decline survives inside firm-product-arm cells that ship in both periods. The fall is not markup compression: the quantity response falls short of what the price cut implies by exactly the amount the measurement identity requires. We then show that this class of evidence cannot go further. Measured quality is a

linear combination of the price and quantity it is inferred from, so a degraded product and a widened clientele imply the same three coefficients. We set out what would separate them.

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake. *Presented at the Midwest Economic Theory and International Trade Meetings, Virginia Tech, 2025.* [\[paper\]](#)

Abstract: This paper extends the Trade Restrictiveness Index (TRI) to preferential tariffs. Using data for 2000–2020, a period of rapid expansion of Preferential Trade Agreements (PTAs), we compute 2,406 annual TRIs for 136 countries in an Armington model with large countries. Once terms-of-trade effects are taken into account, the global import-weighted TRI lies well below the average tariff, and the TRI for preferential tariffs declines faster than the TRI for MFN tariffs as the share of trade under PTAs grows. Across countries, average tariffs track the TRI poorly: the largest importers have TRIs far below their average tariffs, while many small economies have TRIs above them. Assuming that countries are small overstates the TRI, most of all for large importers.

Price Risk, Irreversible Investment, and the Welfare Benchmark for the Section 45Z Clean Fuel Production Credit with T. Edward Yu. [\[paper\]](#) [\[slides\]](#)

How Should 45Z Pay for Risk? A Welfare-CVaR and Project-Finance Decomposition of the IRA Clean Fuel Production Credit with T. Edward Yu. [\[paper\]](#) [\[slides\]](#)

Estimating Pork Primal Price Relationships Using a Threshold Vector Autoregressive Analysis with Charles Martinez, Christopher N. Boyer, T. Edward Yu, and David Anderson. *SSRN working paper 4121953 (2022).* [\[paper\]](#)

A Mean-Field Game of Mayoral Endorsements: Identification and Evidence from Brazilian Coalitional Presidentialism with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).* [\[paper\]](#) [\[slides\]](#)

Abstract: A mayor's return from endorsing a presidential candidate falls as other mayors endorse the same one, so endorsement is a strategic discrete-choice game that has gone structurally unestimated: among Brazil's 5,570 mayors the joint action space is too large to enumerate. Anonymity collapses the problem. When payoffs depend on rivals only through aggregate shares, equilibrium is a fixed point on the candidate simplex whose dimension does not grow with the number of players, and the model estimates by least squares in milliseconds. Estimating it on Brazil's 2014 election, we then show what the exercise does not buy: the prize parameter is algebraically the average log-odds of alignment rescaled by the share of mayors classified as aligned, so it changes sign with the classification rule. What survives is reduced form. A close-race difference-in-differences design puts discretionary transfers to co-partisan mayors 10.6 percent above those of other coalition members and finds no premium on the formula-based transfer that leaves no discretion to allocate.

WORK IN PROGRESS

A Weight-Free Benchmark for Multilateral Tariff Bargaining with Georg Schaur. *Draft, August 2026.* [\[paper\]](#) [\[slides\]](#)

RESEARCH EXPERIENCE

Graduate Research Assistant, University of Tennessee 2021 – 2023
USDA NIFA (two projects) and FAA Office of Environment and Energy (ASCENT) funded research on forest-residue biofuel and sustainable aviation fuel (SAF) supply chains in the Southeast U.S.

- Designed cost-minimizing and profit-maximizing supply-chain networks with mixed-integer programming (GAMS); results published in *Biomass and Bioenergy* and *Transportation Research Record*.

- Facility-siting and investment analysis combining econometric models with the IMPLAN regional impact platform; efficiency analysis of facilities and airports with Data Envelopment Analysis.
- Sensitivity and scenario analysis of feedstock quality and market conditions with extended mathematical programming (EMP), supervised machine learning, and @RISK.
- Managed high-resolution geospatial data in ArcGIS, R, and Python.

Other research experience

- *Panel and causal methods*: synthetic-control estimate of the long-run output effect of the 2008 Wenchuan earthquake; GMM estimates of MFP/CFAP payments on U.S. agricultural lending; Hansen panel-threshold models of green innovation and CO₂; mediation analysis of digital finance and emissions for Chinese listed firms.
- *Time series*: threshold VAR of regime switching in commodity-price linkages; VECM and multivariate GARCH of wholesale commodity-price volatility.
- *Machine learning*: drivers of CO₂ emissions from patent data; equity-premium prediction from *NYT* and *WSJ* text.

PRESENTATIONS

Western Regional Science Association , Annual Meeting, Hawaii	2023
Western Agricultural Economics Association , Annual Meeting, Santa Fe, NM	2022
Southern Agricultural Economics Association , Annual Meeting, New Orleans, LA	2022
Midwest Economic Theory and International Trade Meetings , Virginia Tech	Spring 2025
University of Tennessee , Economics Brown Bag Seminar Series	Spring 2024, Fall 2024
University of Tennessee , Applied Micro Group	Spring 2024, Fall 2024

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics	Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development	Summer 2025
ECON 313: Intermediate Macroeconomics	Summer 2026

Teaching Assistant

Ph.D. courses:

ECON 511: Microeconomic Theory	Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics	Spring 2024
ECON 581: Mathematical Methods in Economics	Summer 2024

Master's courses:

AREC 525: Agribusiness Operations Research Methods	Spring 2022, Spring 2023
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HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching, University of Tennessee	2026
Graduate Summer Fellowship, University of Tennessee	2025
J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present
Graduate Assistantship, University of Tennessee	2021 – present
Outstanding Rainbow Volunteer, Tongji University (tutoring pupils in Shanghai)	2013 – 2015

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.

Methods: panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.

Credentials: CFA Level II.

Languages: Chinese (native), English (fluent).

REFERENCES

T. Edward Yu

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Georg Schaur

Leigh Burch III Economics Professor and Haslam Family Faculty Fellow

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